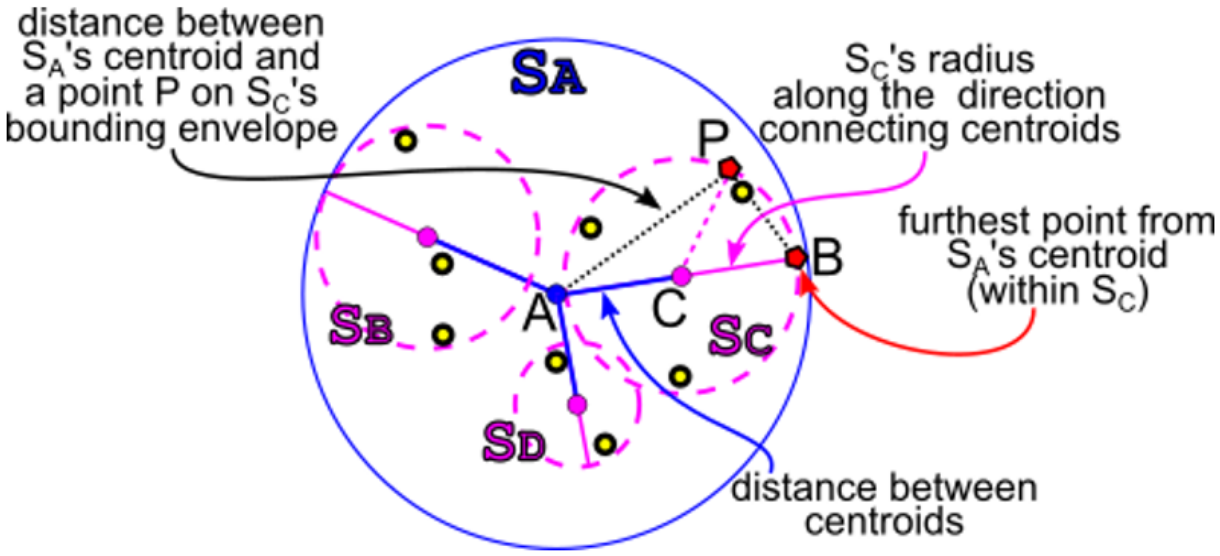


Figure 10.17 Computing the radius of an internal node: the point in S_C further away from S_A's centroid A is the point on the bounding envelope that's further in the opposite direction WRT S_A's centroid, and its distance is therefore the sum of the distance A-C between the two centroids, plus S_C's radius. If we choose another point P on the bounding envelope, its distance from A must be smaller than the distance A-B, because metrics by definition needs to obey the triangular inequality, and the other two edges of triangle ACP are AC and CP, which is S_C's radius. You can check this is also true for any of the other envelopes in the figure.

This method computes the centroid for a node, as the center of mass of its children's centroids: remember, for leaves, their children are just the points it contains, while for internal nodes, their children are other nodes.